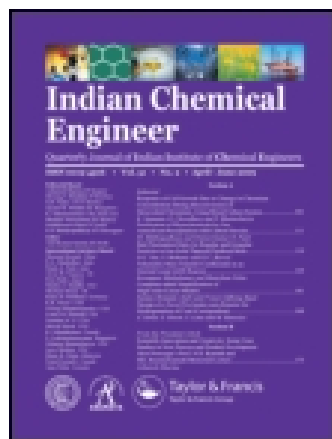




## Indian Chemical Engineer

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/tice20>

### Analytical Expressions for the Adsorbate Breakthrough Curve from a Fixed Bed of Adsorbent with First-order and Second-order Kinetic Models

Sean Skrip<sup>a</sup>, Gautam K. Das<sup>b</sup> & Siddharth G. Chatterjee<sup>c</sup>

<sup>a</sup> Division of Environmental Science, SUNY College of Environmental Science and Forestry, 1 Forestry Drive, Syracuse, NY, 13210, USA

<sup>b</sup> 118/A Acharya Jagdish Chandra Bose Road, Kolkata, 700014, India

<sup>c</sup> Department of Paper and Bioprocess Engineering, SUNY College of Environmental Science and Forestry, 1 Forestry Drive, Syracuse, NY, 13210, USA

Published online: 06 Jun 2013.

To cite this article: Sean Skrip, Gautam K. Das & Siddharth G. Chatterjee (2013) Analytical Expressions for the Adsorbate Breakthrough Curve from a Fixed Bed of Adsorbent with First-order and Second-order Kinetic Models, Indian Chemical Engineer, 55:2, 87-103, DOI: [10.1080/00194506.2013.798490](https://doi.org/10.1080/00194506.2013.798490)

To link to this article: <http://dx.doi.org/10.1080/00194506.2013.798490>

PLEASE SCROLL DOWN FOR ARTICLE

Taylor & Francis makes every effort to ensure the accuracy of all the information (the "Content") contained in the publications on our platform. However, Taylor & Francis, our agents, and our licensors make no representations or warranties whatsoever as to the accuracy, completeness, or suitability for any purpose of the Content. Any opinions and views expressed in this publication are the opinions and views of the authors, and are not the views of or endorsed by Taylor & Francis. The accuracy of the Content should not be relied upon and should be independently verified with primary sources of information. Taylor and Francis shall not be liable for any losses, actions, claims, proceedings, demands, costs, expenses, damages, and other liabilities whatsoever or howsoever caused arising directly or indirectly in connection with, in relation to or arising out of the use of the Content.

This article may be used for research, teaching, and private study purposes. Any substantial or systematic reproduction, redistribution, reselling, loan, sub-licensing, systematic supply, or

distribution in any form to anyone is expressly forbidden. Terms & Conditions of access and use can be found at <http://www.tandfonline.com/page/terms-and-conditions>



Taylor & Francis

Taylor & Francis Group

# Analytical Expressions for the Adsorbate Breakthrough Curve from a Fixed Bed of Adsorbent with First-order and Second-order Kinetic Models

Sean Skrip<sup>1</sup>, Gautam K. Das<sup>2</sup> & Siddharth G. Chatterjee<sup>3\*</sup>

<sup>1</sup>Division of Environmental Science, SUNY College of Environmental Science and Forestry, 1 Forestry Drive, Syracuse, NY 13210, USA

<sup>2</sup>118/A Acharya Jagdish Chandra Bose Road, Kolkata 700014, India

<sup>3</sup>Department of Paper and Bioprocess Engineering, SUNY College of Environmental Science and Forestry, 1 Forestry Drive, Syracuse, NY 13210, USA

**Abstract:** In many batch studies, the kinetics of adsorption have demonstrated adherence to first-order (Lagergren) or second-order (Ho and McKay) models. Since practical adsorption systems generally operate in the fixed-bed mode, in this work, the first- and second-order kinetic models describing adsorbate uptake are combined with an overall adsorbate material balance to obtain simple analytical expressions for the adsorbate breakthrough curve from a fixed bed of adsorbent. These expressions can correlate some fixed-bed experimental data reported in the literature with an average error of 18%. The equations describing the breakthrough behaviour can be used for the rapid scale-up and design of fixed-bed adsorption columns for removing contaminants from water and wastewater.

**Keywords:** Breakthrough curve, Fixed-bed adsorption, Ho and McKay model, Lagergren model, Water treatment

## Introduction

Modern industrial processes produce a variety of hazardous materials like metal ions, dyes and organic compounds, which have to be removed from waste streams before their discharge into the environment. The removal of such chemicals from liquid and gaseous effluent streams by carbon

---

\*Author for Correspondence. E-mail: [schatterjee@esf.edu](mailto:schatterjee@esf.edu)

adsorption, in order to comply with environmental regulations, has been a long-standing practice. Many novel and low-cost adsorbents have been tested for removal of hazardous chemicals [1–23]. Examples of these alternative adsorbents are natural materials like cedar sawdust, popcorn, kudzu, sphagnum moss peat, carbons prepared from tamarind nutshell, rice husk and coconut husk [3–9], as well as industrial by-products or waste products like crushed brick, mustard-oil cake, activated slag created in steel blast furnaces, red mud resulting from aluminium processing, and even carbon developed from waste slurry generated in fertilizer plants [9–15]. These works have investigated the effects of temperature, initial pH, initial concentration and adsorbent particle size on the adsorption process, and have also presented Langmuir and Freundlich isotherms in order to determine sorbent dosage requirements. In many of these studies, which were conducted in batch systems, the adsorbate uptake was found to obey the first-order model of Lagergren [24] or the second-order model of Ho and McKay [25]. Assuming no mass-transfer effects, Azizian [26] has provided a theoretical justification for the first- and second-order kinetic models in the case of batch adsorption.

Adsorption systems generally operate in the fixed-bed mode for which sophisticated design procedures exist [27]. From a practical point of view, such procedures involve complex mathematical and numerical techniques and presume a high level of technical competence and skill on the part of the practitioner, as well as the associated computational facilities necessary to implement the techniques. These are often lacking in the villages and towns of the countries of Latin America, Africa and Asia, whose populations are increasingly threatened with contamination of their drinking water supplies owing to rapid industrialization and natural factors (e.g. groundwater contamination by geogenic arsenic in West Bengal, India and Bangladesh).

The main objective of this paper is to present simple and alternative methods for the scale-up and design of fixed-bed adsorption columns for removing contaminants (e.g. arsenic, iron, chromium and organic dyes) from water and wastewater, which can be easily implemented without requiring significant expertise or computational facilities. Specifically, the first- and second-order kinetic models describing adsorbate uptake [24, 25], which were originally developed for batch adsorption, are combined with an overall adsorbate material balance in order to derive analytical expressions for the adsorbate breakthrough curve from a fixed bed of adsorbent. These expressions are used to correlate some fixed-bed experimental data that have been reported in the literature.

## Theory

Consider a fixed bed of length  $L$  containing an adsorbent. A fluid, which has an adsorbable species with a concentration of  $c_{\text{in}}$ , enters the bed at  $x=0$  with a superficial velocity  $u_s$ , and leaves it with a concentration of  $c_{\text{eff}}$  at  $x=L$ . The problem is to predict the behaviour of  $c_{\text{eff}}$  as a function of the elapsed time  $t$  since the start of the process at  $t=0$  (i.e. the breakthrough curve). Neglecting axial dispersion compared with convective transport, the differential mass balance for the adsorbate in the bed is given by

$$\varepsilon_b \frac{\partial c}{\partial t} + u_s \frac{\partial c}{\partial x} + \rho_b \frac{\partial q}{\partial t} = 0 \quad (1)$$

Here,  $\varepsilon_b$  and  $\rho_b$  are the porosity and bulk density of the bed, respectively, whereas  $c$  and  $q$  are the fluid-phase and solid-phase concentrations, respectively, of the adsorbate at a distance  $x$  from the inlet of the bed. The initial and boundary conditions for Equation (1) are

$$c = 0, q = 0 \text{ at } t = 0, \text{ for } x \geq 0 \quad (2)$$

$$c = c_{\text{in}}, \text{ for } t > 0, \text{ at } x = 0 \quad (3)$$

Integrating Equation (1) over the bed from  $x=0$  (where  $c=c_{\text{in}}$ ) to  $x=L$  (where  $c=c_{\text{eff}}$ ), and noting that  $\rho_b q \gg \varepsilon_b c$ , we obtain

$$\frac{c_{\text{eff}}}{c_{\text{in}}} = 1 - \frac{\rho_b L}{u_s c_{\text{in}}} \frac{d\bar{q}}{dt} \quad (4)$$

where  $\bar{q}$  is the average solid-phase adsorbate concentration in the bed given by

$$\bar{q} = \frac{1}{L} \int_0^L q dx \quad (5)$$

### First-order Model

The first-order kinetic model, which has been empirically verified in numerous batch-adsorption studies, is given by [24, 25]

$$\frac{\partial q}{\partial t} = k_1 (q_e - q) \quad (6)$$

where  $k_1$  is a first-order rate constant, which includes the effects of mass transfer in the fluid and solid phases, physical adsorption and/or chemical reaction, and  $q_e$  is an equilibrium solid-phase concentration of adsorbate, which, in the case of batch adsorption, has been typically taken to be the long-time or ultimate concentration of the adsorbate in the solid when it has achieved equilibrium with the fluid. In this paper, which is concerned with fixed-bed adsorption, we will assume that  $k_1$  and  $q_e$  (adsorbent sorption capacity) are constants that have to be determined empirically. A theoretical justification for Equation (6), which includes mass-transfer effects, is provided in the appendix in the case of physical adsorption.

Integrating Equation (6) over the bed from  $x=0$  to  $x=L$ , we obtain

$$\frac{d\bar{q}}{dt} = k_1 (q_e - \bar{q}) \quad (7)$$

whose solution, subject to the initial condition  $\bar{q} = 0$  at  $t=0$ , is

$$\bar{q} = q_e [1 - \exp(-k_1 t)] \quad (8)$$

Using Equation (8) in Equation (4) yields

$$\frac{c_{\text{eff}}}{c_{\text{in}}} = 1 - \frac{\rho_b L k_1 q_e}{u_s c_{\text{in}}} \exp(-k_1 t) \quad (9)$$

Let  $n$  be the number of bed volumes of fluid that have passed through the bed since  $t=0$ , that is,

$$n = \frac{u_s(t - L\varepsilon_b/u_s)}{L} = \frac{Qt}{V_{\text{bed}}} - \varepsilon_b \approx \frac{Qt}{V_{\text{bed}}} \quad (10)$$

where  $Q$  and  $V_{\text{bed}}$  are the volumetric flow rate of the fluid through the adsorbent bed and the volume of the bed, respectively. The expression for the breakthrough curve given by Equation (9) is then transformed into

$$\frac{c_{\text{eff}}}{c_{\text{in}}} = 1 - Da_1 n_{\text{stoic}} \exp(-nDa_1) \quad (11a)$$

or

$$\frac{c_{\text{eff}}}{c_{\text{in}}} = 1 - \exp[-Da_1(n - n_{\text{BT}})] \quad (11b)$$

where  $n_{\text{BT}}$  is the number of bed volumes at breakthrough (i.e. when  $c_{\text{eff}}=0$ ). The dimensionless parameters  $Da_1$  and  $n_{\text{stoic}}$  are given by

$$Da_1 = \frac{k_1 L}{u_s} = \frac{k_1 V_{\text{bed}}}{Q} \quad (12)$$

$$n_{\text{stoic}} = \frac{\rho_b q_e}{c_{\text{in}}} \quad (13)$$

Thus, the two parameters that govern breakthrough are either  $Da_1$  and  $n_{\text{stoic}}$  [Equation (11a)], or  $Da_1$  and  $n_{\text{BT}}$  [Equation (11b)]. Equating Equations (11a) and (11b), we obtain

$$n_{\text{BT}} = \frac{\ln(Da_1 n_{\text{stoic}})}{Da_1} \quad (14)$$

which, upon differentiating with respect to  $Da_1$ , assuming  $n_{\text{stoic}}$  to be constant, and setting the derivative equal to zero, yields the maximum number of bed volumes at breakthrough as

$$n_{\text{BT,max}} = \frac{n_{\text{stoic}}}{e} = 0.368 n_{\text{stoic}} \quad (15)$$

Equation (11b) can also be written as

$$\ln\left(\frac{1}{1 - c_{\text{eff}}/c_{\text{in}}}\right) = Da_1(n - n_{\text{BT}}) \quad (16)$$

Hence, if the adsorbate uptake follows a first-order process (i.e. Equation (6)), plotting effluent breakthrough data according to Equation (16) will allow one to determine the two parameters  $Da_1$  and  $n_{\text{BT}}$  of the breakthrough curve.

## Second-order Model

Several adsorbate–adsorbent systems have been found to obey the second-order kinetic model of Ho and McKay [25], which is given by

$$\frac{\partial q}{\partial t} = k_2(q_e - q)^2 \quad (17)$$

where  $k_2$  is a second-order rate constant, which includes the effects of mass transfer, physical adsorption and/or chemical reaction. Integrating Equation (17) over the bed from  $x=0$  to  $x=L$ , and assuming that

$$\overline{(q_e - q)^2} \approx (\overline{q_e - q})^2 = (q_e - \bar{q})^2$$

we get

$$\frac{d\bar{q}}{dt} = k_2(q_e - \bar{q})^2 \quad (18)$$

whose solution, subject to the initial condition of  $\bar{q} = 0$  at  $t=0$ , is given by

$$\bar{q} = \frac{q_e^2 k_2 t}{1 + q_e k_2 t} \quad (19)$$

Using Equation (19) in Equation (4) yields

$$\frac{c_{\text{eff}}}{c_{\text{in}}} = 1 - \frac{\rho_b L k_2 q_e^2}{u_s c_{\text{in}} (1 + k_2 q_e t)^2} \quad (20)$$

Equation (20) can also be expressed as

$$\frac{c_{\text{eff}}}{c_{\text{in}}} = 1 - \frac{n_{\text{stoic}} Da_2}{(1 + n Da_2)^2} \quad (21a)$$

or

$$\frac{c_{\text{eff}}}{c_{\text{in}}} = 1 - \left( \frac{1 + n_{\text{BT}} Da_2}{1 + n Da_2} \right)^2 \quad (21b)$$

where

$$Da_2 = \frac{q_e k_2 L}{u_s} = \frac{q_e k_2 V_{\text{bed}}}{Q} \quad (22)$$

The two parameters governing breakthrough are thus either  $Da_2$  and  $n_{\text{stoic}}$  [Equation (21a)], or  $Da_2$  and  $n_{\text{BT}}$  [Equation (21b)]. Equating Equations (21a) and (21b), we obtain

$$n_{\text{BT}} = \frac{\sqrt{n_{\text{stoic}} Da_2} - 1}{Da_2} \quad (23)$$

Following a similar procedure as for the first-order model yields  $n_{\text{BT,max}}$  as

$$n_{\text{BT,max}} = \frac{n_{\text{stoic}}}{4} = 0.25 n_{\text{stoic}} \quad (24)$$

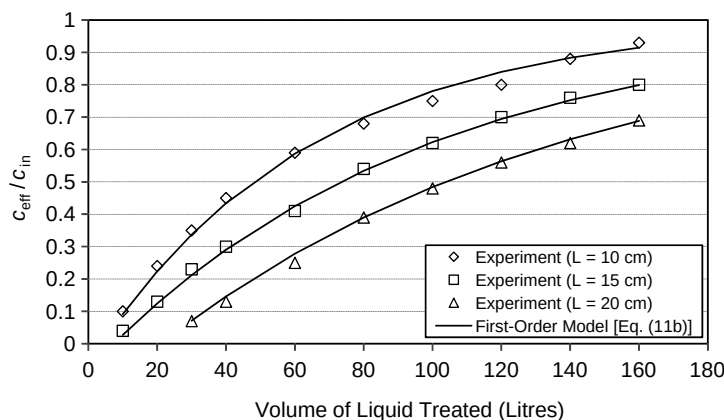
Equation (21b) can also be written as

$$\ln \sqrt{1 - c_{\text{eff}}/c_{\text{in}}} = \ln(1 + n_{\text{BT}} Da_2) - \ln(1 + n Da_2) \quad (25)$$

which will allow the determination of the two parameters  $Da_2$  and  $n_{\text{BT}}$  of the breakthrough curve in the case of second-order behaviour.

## Results and Discussion

The models were correlated against experimental data from three separate investigations [1–3] that involved the adsorption of a component from a liquid or gaseous phase on a solid adsorbent. In order to discriminate between first- and second-order behaviour, the data were first plotted according to Equation (16). If the correlation coefficient of the fit was  $>0.98$ , first-order behaviour was presumed; if this was not the case, we fitted the data with the second-order expression given by Equation (21b). Walker and Weatherley [1], who used the bed-depth-service-time model to correlate their data, studied the adsorption of acid dyes in fixed beds of granular activated carbon. Figure 1 shows their experimental breakthrough data of an acid dye (Tectilon Red 2B) at bed depths of 10, 15 and 20 cm, respectively; we observe that the data conform closely to the first-order expression [Equation (11b)]. The values of the relevant experimental parameters are provided in Table 1, and the values of  $q_e$  and  $k_1$  obtained from the fit are reported in Table 4. The value of the adsorbent capacity ( $q_e$ ) is fairly consistent; it is 0.0575 g/g on average whereas the value of  $k_1$  decreases by 33 and 20% as the bed depth increases by 50 and 33%, respectively (Table 4). From theoretical considerations, a correlation for  $k_1$  as a function of particle, bed and fluid properties is



**Fig. 1.** Comparison of experimental [1] and theoretical breakthrough curves in the adsorption of an acid dye (Tectilon Red 2B) on granular activated carbon – effect of bed depth. Experimental and kinetic parameters are given in Tables 1 and 4, respectively.

**Table 1.** Experimental parameters of Fig. 1 (from Walker and Weatherley [1])

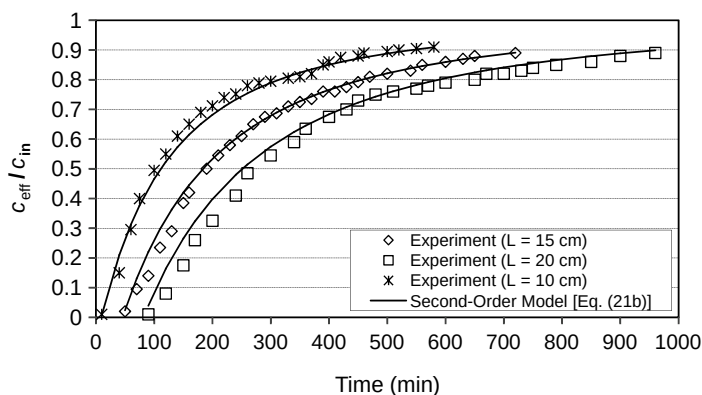
| Parameter                               | Value                    |
|-----------------------------------------|--------------------------|
| Depth of adsorbent bed, $L$             | 10, 15 and 20 cm         |
| Diameter of adsorbent bed, $d$          | 5 cm                     |
| Inlet adsorbate concentration, $c_{in}$ | 0.0001 g/cm <sup>3</sup> |
| Volumetric flow rate of liquid, $Q$     | 50 cm <sup>3</sup> /min  |
| Bulk density of adsorbent bed, $\rho_b$ | 0.611 g/cm <sup>3</sup>  |
| Adsorbent particle size, $d_p$          | 0.0355–0.05 cm           |

proposed in the appendix [Equation (A12)], which suggests that, for liquids flowing through a bed of adsorbent particles,

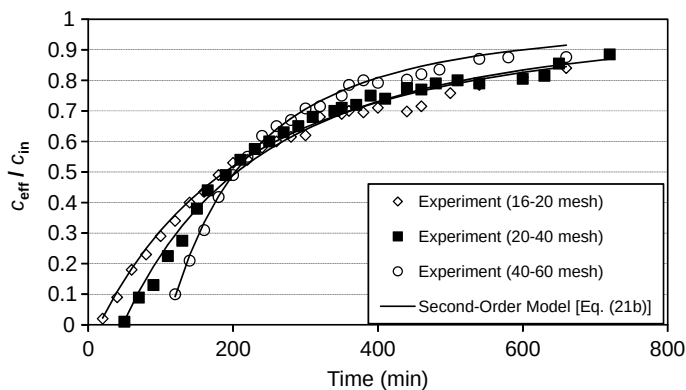
$$k_1 = A \frac{u_s}{\varepsilon_b} \left( \frac{n_{stoic}}{\varepsilon_b} + 1 \right)^B \frac{1}{L^{2-C} d_p^{C-1}} \quad (26)$$

where  $A$ ,  $B$  and  $C$  are empirical constants with  $d_p$  being the (average) diameter of the adsorbent particles. The value of  $C$  for the data in Fig. 1 is 1.084.

Figures 2–4 show breakthrough data from the work of Han et al. [2], who studied the individual and competitive removal of copper (II) and lead (II) from aqueous solution in fixed beds of manganese oxide coated zeolite; they used the Thomas model to represent breakthrough behaviour. Overall, the data for copper show a good match to the second-order model [Equation (21b)]; values of  $q_e$  and  $k_2$  obtained from the fit are given in Table 4, while Table 2 reports values of the experimental parameters. For the data in Fig. 2, the value of  $q_e$  is nearly constant at 0.0122 g/g, but  $k_2$  decreases by 17 and 12% as the bed depth increases by 50 and 33%, respectively. If one assumes that Equation (26) also describes the behaviour of the second-order rate constant  $k_2$ ,

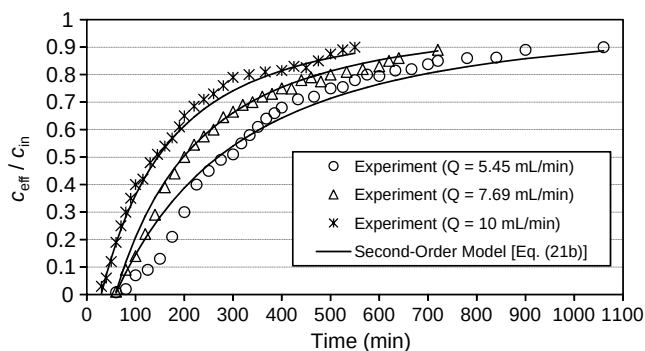


**Fig. 2.** Comparison of experimental (figure 1(a) in Han et al. [2]) and theoretical breakthrough curves in the adsorption of copper on manganese oxide coated zeolite – effect of bed depth. Experimental and kinetic parameters are given in Tables 2 and 4, respectively.



**Fig. 3.** Comparison of experimental (figure 4(a) in Han et al. [2]) and theoretical breakthrough curves in the adsorption of copper on manganese oxide coated zeolite – effect of adsorbent particle size. Experimental and kinetic parameters are given in Tables 2 and 4, respectively.

$C = 1.539$  for the data shown in Fig. 2. Decreasing the adsorbent particle size from 16–20 mesh to 40–60 mesh increases the value of  $k_2$  by a factor of 3.4 owing to increased particle specific surface area, with consequent steepening in breakthrough behaviour and crossover of the breakthrough curves (Fig. 3 and Table 4). This effect of a smaller particle size on improved column performance (i.e. later breakthrough) was also observed by Walker and Weatherley [1], who attributed it to higher interfacial area and increased sorption capacity of the adsorbent. Increasing the rate of liquid flow  $Q$  through the bed from 5.45 to 10 ml/min increases  $k_2$  by 80% owing to enhanced mass transfer (Fig. 4 and Table 4), causing a steepening of the breakthrough curve and earlier breakthrough; this latter observation was also made by Walker and Weatherley [1]. For the data in Fig. 4,  $k_2$  is proportional to  $Q^{0.991}$ , which conforms very well to the linear relationship between the rate constant and superficial liquid velocity as predicted by Equation (26), assuming, as mentioned



**Fig. 4.** Comparison of experimental (figure 2(a) in Han et al. [2]) and theoretical breakthrough curves in the adsorption of copper on manganese oxide coated zeolite – effect of liquid flow rate. Experimental and kinetic parameters are given in Tables 2 and 4, respectively.

**Table 2.** Experimental parameters of Figs 2–4 (from Han et al. [2])

| Parameter                                      | Value                                                               |
|------------------------------------------------|---------------------------------------------------------------------|
| Depth of adsorbent bed, $L$                    | 10 (Fig. 2), 15 (Figs 2–4) and 20 cm (Fig. 2)                       |
| Diameter of adsorbent bed, $d$                 | 1 cm                                                                |
| Inlet adsorbate concentration, $c_{\text{in}}$ | 0.976 mmol/L ( $6.202 \times 10^{-5}$ g/cm <sup>3</sup> )           |
| Volumetric flow rate of liquid, $Q$            | 5.45 (Fig. 4), 7.69 (Figs 2–4) and 10 cm <sup>3</sup> /min (Fig. 4) |
| Bulk density of adsorbent bed, $\rho_b$        | 1.273 g/cm <sup>3</sup>                                             |
| Adsorbent particle size                        | 16–20 mesh (Fig. 3), 20–40 mesh (Figs 2–4) and 40–60 mesh (Fig. 3)  |
| Molecular weight of adsorbate (Cu), MW         | 63.55 g/mol                                                         |
| Experimental temperature, $T$                  | 293 K                                                               |

earlier, that this type of equation is also valid for  $k_2$ . For the experimental data shown in Figs 2–4, the value of the adsorbent capacity ( $q_e$ ) is quite consistent; it is 0.013 g/g on average (Table 4).

Figure 5 shows the rather good fit of the second-order model to the experimental breakthrough data of Mak et al. [3], who studied the adsorption of trichloroethylene (TCE) from an air stream in a fixed bed of popcorn (Table 3). These authors determined values of the sorption rate constant and the equilibrium sorption capacity (for acetone, TCE and toluene), which are analogous to  $k_1$  and  $q_e$  in our work, by fitting the Wheeler or Mecklenburg equation to the experimental 10% breakthrough time (i.e. when  $c_{\text{eff}}/c_{\text{in}} = 0.1$ ). The data shown in Fig. 5 are for an inlet concentration of TCE of 7000 ppm; the fit of the second-order model to their breakthrough data did not reach this level for inlet concentrations of 3000 and 5000 ppm, presumably because the adsorbent bed was far from saturation. We were also unable to get good agreement of the first- and second-order models with steep S-shaped experimental breakthrough curves that have been reported for some systems [9–12]. Thus, the first-order and second-order expressions for the breakthrough curve [Equations (11b) and (21b)] have the ability to represent convex-shaped breakthrough data (such as those shown in Figs 1–5) quite well. The average root-mean-square deviation between prediction and experiment for all the cases presented in Figs 1–5 is 18% (Table 4).

Values of the dimensionless parameters of the first- and second-order kinetic models for the data in Figs 1–4 are gathered together in Table 5. The last column of this table shows values of  $n_{\text{BT, max}}$ ,

**Table 3.** Experimental parameters of Fig. 5 (from Mak et al. [3])

| Parameter                                      | Value                                                 |
|------------------------------------------------|-------------------------------------------------------|
| Volume of adsorbent bed, $V_{\text{bed}}$      | 450 cm <sup>3</sup>                                   |
| Inlet adsorbate concentration, $c_{\text{in}}$ | 7000 ppm ( $3.784 \times 10^{-5}$ g/cm <sup>3</sup> ) |
| Volumetric flow rate of fluid, $Q$             | 100 cm <sup>3</sup> /min                              |
| Bulk density of adsorbent bed, $\rho_b$        | 0.045 g/cm <sup>3</sup>                               |
| Specific surface area of adsorbent, $S_p$      | 30,000 cm <sup>2</sup> /g                             |
| Molecular weight of adsorbate (TCE), MW        | 131.35 g/mol                                          |
| Experimental temperature, $T$                  | 296.15 K                                              |

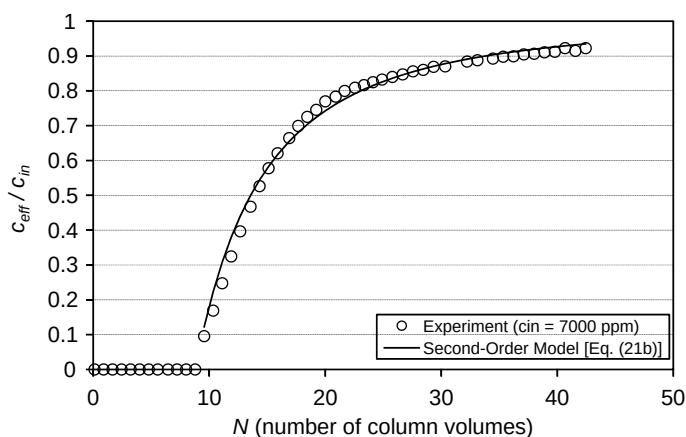
TCE, Trichloroethylene.

**Table 4.** Values of kinetic parameters of the first- and second-order models and the root-mean-square (rms) error of prediction in Figs 1–5

| Figure no.             | $q_e$ (g/g) | $k_1$ (1/min)          | $k_2$ (1/min) | rms Error (%)     |
|------------------------|-------------|------------------------|---------------|-------------------|
| 1 ( $L = 10$ cm)       | 0.0562      | $7.894 \times 10^{-4}$ |               | 4.4               |
| 1 ( $L = 15$ cm)       | 0.0570      | $5.263 \times 10^{-4}$ |               | 10.5              |
| 1 ( $L = 20$ cm)       | 0.0593      | $4.202 \times 10^{-4}$ |               | 6.0               |
| 2 ( $L = 10$ cm)       | 0.0122      |                        | 0.344         | 10.7              |
| 2 ( $L = 15$ cm)       | 0.0123      |                        | 0.285         | 15.0              |
| 2 ( $L = 20$ cm)       | 0.0122      |                        | 0.25          | 61.7 <sup>a</sup> |
| 3 (16–20 mesh)         | 0.0138      |                        | 0.179         | 5.7               |
| 3 (20–40 mesh)         | 0.0138      |                        | 0.217         | 12.6              |
| 3 (40–60 mesh)         | 0.0139      |                        | 0.612         | 3.7               |
| 4 ( $Q = 5.45$ ml/min) | 0.0129      |                        | 0.171         | 66.4 <sup>b</sup> |
| 4 ( $Q = 7.69$ ml/min) | 0.0129      |                        | 0.277         | 12.9              |
| 4 ( $Q = 10$ ml/min)   | 0.0129      |                        | 0.308         | 17.1              |
| 5                      | 0.0405      |                        | 1.974         | 8.8               |

<sup>a</sup> Decreases to 13% if the first two data points are neglected. <sup>b</sup> Decreases to 13.7% if the first five data points are neglected.

which were obtained from either Equation (15) or Equation (24) using the average values of  $n_{\text{stoic}}$  reported in the second column of the table. It can be concluded that, except for the experimental run with the 40–60 mesh particle size (Fig. 3), all other column runs in Figs 1–4 were far from optimum, that is, they had significantly smaller  $n_{\text{BT}}$  values than the (relevant) theoretical limit,  $n_{\text{BT,max}}$ .

**Fig. 5.** Comparison of experimental [3] and theoretical breakthrough curves in the adsorption of trichloroethylene on popcorn. Experimental and kinetic parameters are given in Tables 3 and 4, respectively.

**Table 5.** Values of dimensionless parameters of the first- and second-order models in Figs 1–4

| Figure no.             | $n_{\text{stoic}}$ | $Da_1$                | $Da_2$                | $n_{\text{BT}}$ | $n_{\text{BT,max}}$ |
|------------------------|--------------------|-----------------------|-----------------------|-----------------|---------------------|
| 1 ( $L = 10$ cm)       | 343.38             | $3.10 \times 10^{-3}$ |                       | 20.15           |                     |
| 1 ( $L = 15$ cm)       | 348.27             | $3.10 \times 10^{-3}$ |                       | 24.73           |                     |
| 1 ( $L = 20$ cm)       | 362.32             | $3.30 \times 10^{-3}$ |                       | 54.17           |                     |
| Average                | 351.33             |                       |                       |                 | 129.25              |
| 2 ( $L = 10$ cm)       | 250.41             |                       | $4.29 \times 10^{-3}$ | 8.40            |                     |
| 2 ( $L = 15$ cm)       | 252.47             |                       | $5.37 \times 10^{-3}$ | 30.61           |                     |
| 2 ( $L = 20$ cm)       | 250.41             |                       | $6.23 \times 10^{-3}$ | 39.97           |                     |
| Average                | 251.10             |                       |                       |                 | 62.77               |
| 3 (16–20 mesh)         | 283.25             |                       | $3.78 \times 10^{-3}$ | 9.34            |                     |
| 3 (20–40 mesh)         | 283.25             |                       | $4.59 \times 10^{-3}$ | 30.50           |                     |
| 3 (40–60 mesh)         | 285.31             |                       | $1.30 \times 10^{-3}$ | 71.23           |                     |
| Average                | 283.94             |                       |                       |                 | 70.98               |
| 4 ( $Q = 5.45$ ml/min) | 264.78             |                       | $4.77 \times 10^{-3}$ | 25.93           |                     |
| 4 ( $Q = 7.69$ ml/min) | 264.78             |                       | $5.47 \times 10^{-3}$ | 37.25           |                     |
| 4 ( $Q = 10$ ml/min)   | 264.78             |                       | $4.68 \times 10^{-3}$ | 24.20           |                     |
| Average                | 264.78             |                       |                       |                 | 66.20               |

Metal-ion adsorption is generally pH-dependent [27–32]. Since the models presented in this work are concerned with adsorption of a single species, they are unable to account for effects of the solution pH. However, it may be possible to include pH effects in the kinetic approach by developing two-species or three-species models of competitive adsorption. From the point of view of practical application, this can be a simple route to designing fixed-bed adsorption columns for the removal of iron and arsenic, which are often present together from groundwater. As can be seen from Equations (11b) and (21b), only two parameters are needed to describe the breakthrough curve in the kinetic method, both of which can be directly estimated from experimental breakthrough data with no subsidiary experiments (e.g. batch studies for determining sorption equilibria and intraparticle diffusion coefficients) being necessary. While this lumped-parameter approach may not be as technically sophisticated as the distributed-parameter or mechanistic approach practised in the adsorption field [27], it nevertheless has the qualities of harmony and simplicity, and allows one to grasp the totality of the fixed-bed adsorption process.

## Conclusion

First- and second-order kinetic models of adsorbate uptake, which have been used extensively in the literature to correlate results of batch adsorption experiments, were combined with an overall adsorbate material balance to derive analytical expressions for adsorbate breakthrough in a fixed bed of adsorbent. These expressions correlated some convex-shaped experimental breakthrough data, reported in the literature, with an average error of 18%. The simple, analytical expressions for the breakthrough curve presented in this paper can guide the adsorption practitioner in developing rapid scale-up and design methods for fixed-bed adsorption columns in the field from experimental data obtained with a laboratory or pilot column.

## Nomenclature

|                      |                                                                                                                                                   |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| $a_p$                | Radius of an adsorbent particle, cm                                                                                                               |
| $A$                  | Empirical constant in Equation (A12)                                                                                                              |
| $A_{\text{bed}}$     | Area of cross section of the adsorbent bed ( $\pi d^2/4$ ), $\text{cm}^2$                                                                         |
| $B$                  | Empirical constant in Equation (A12)                                                                                                              |
| $c$                  | Fluid-phase concentration of the adsorbate at any location $x$ in the adsorbent bed or in batch adsorption, $\text{g/cm}^3$                       |
| $C$                  | Empirical constant in Equation (A12)                                                                                                              |
| $C_{\text{ads}}$     | Adsorbent dosage in batch adsorption, $\text{g/cm}^3$                                                                                             |
| $c_{\text{eff}}$     | Effluent fluid-phase concentration of the adsorbate (i.e. at $x=L$ ) in the adsorbent bed, $\text{g/cm}^3$                                        |
| $c_{\text{in}}$      | Inlet fluid-phase concentration of the adsorbate (i.e. at $x=0$ ) in the adsorbent bed, $\text{g/cm}^3$                                           |
| $c_s$                | Fluid-phase concentration of the adsorbate at the surface of an adsorbent particle in batch adsorption, $\text{g/cm}^3$                           |
| $c_0$                | Initial fluid-phase concentration of the adsorbate in batch adsorption, $\text{g/cm}^3$                                                           |
| $d$                  | Diameter of the adsorbent bed, cm                                                                                                                 |
| $D$                  | Molecular diffusion coefficient of the adsorbate in the carrier gas, $\text{cm}^2/\text{min}$                                                     |
| $D_e$                | Effective diffusion coefficient of the adsorbate in an adsorbent particle, $\text{cm}^2/\text{min}$                                               |
| $D_L$                | Axial dispersion coefficient of the adsorbate in the adsorbent bed, $\text{cm}^2/\text{min}$                                                      |
| $d_p$                | Size or diameter of an adsorbent particle, cm                                                                                                     |
| $Da_1$               | Dimensionless first-order rate constant defined by Equation (12)                                                                                  |
| $Da_2$               | Dimensionless second-order rate constant defined by Equation (22)                                                                                 |
| $k_{\text{eff}}$     | Effective or overall mass-transfer coefficient defined by Equation (A9), $1/\text{min}$                                                           |
| $k_f$                | Fluid-phase mass-transfer coefficient in batch adsorption, $\text{cm}/\text{min}$                                                                 |
| $k_f'$               | Defined by Equation (A10), $\text{cm}^3/\text{g min}$                                                                                             |
| $k_p$                | Intraparticle mass-transfer coefficient, $1/\text{min}$                                                                                           |
| $k_1$                | First-order rate constant, $1/\text{min}$                                                                                                         |
| $k_2$                | Second-order rate constant, $1/\text{min}$                                                                                                        |
| $K_{\text{eq}}$      | Slope of adsorption isotherm, $\text{cm}^3/\text{g}$                                                                                              |
| $L$                  | Length or depth of the adsorbent bed, cm                                                                                                          |
| $M$                  | Mass of carbon in the adsorbent bed, g                                                                                                            |
| $MW$                 | Molecular weight of the adsorbate, $\text{g/mol}$                                                                                                 |
| $n$                  | Number of bed volumes of fluid that have passed through the adsorbent bed since $t=0$                                                             |
| $n_{\text{BT}}$      | Number of bed volumes of fluid that have passed through the adsorbent bed at breakthrough (i.e. when $c_{\text{eff}}=0$ )                         |
| $n_{\text{BT, max}}$ | Maximum value of $n_{\text{BT}}$                                                                                                                  |
| $n_{\text{stoic}}$   | Defined by Equation (13)                                                                                                                          |
| $q$                  | Solid-phase concentration of the adsorbate at any location $x$ in the adsorbent bed or in an adsorbent particle in batch adsorption, $\text{g/g}$ |
| $Q$                  | Volumetric flow rate of the fluid in the adsorbent bed, $\text{cm}^3/\text{min}$                                                                  |
| $\bar{q}$            | Average solid-phase concentration of the adsorbate in the adsorbent bed ( $\text{g/g}$ )                                                          |
| $q_e$                | Capacity of the adsorbent for adsorbate, $\text{g/g}$                                                                                             |
| $q_s$                | Solid-phase concentration of the adsorbate at the surface of an adsorbent particle in batch adsorption, $\text{g/g}$                              |
| $q_0$                | Initial solid-phase concentration of the adsorbate in batch adsorption, $\text{g/g}$                                                              |
| $Re_p$               | Particle Reynolds number defined by Equation (A15)                                                                                                |
| $S_p$                | Specific surface area of the adsorbent, $\text{cm}^2/\text{g}$                                                                                    |

|                  |                                                                                 |
|------------------|---------------------------------------------------------------------------------|
| $t$              | Elapsed time, min                                                               |
| $T$              | Experimental temperature, K                                                     |
| $u_s$            | Superficial fluid velocity in the adsorbent bed ( $LQ/V_{\text{bed}}$ ), cm/min |
| $V_{\text{bed}}$ | Volume of the adsorbent bed ( $A_{\text{bed}}L$ ), $\text{cm}^3$                |
| $x$              | Axial location in the adsorbent bed measured from its inlet, cm                 |

## Greek Symbols

|              |                                                                           |
|--------------|---------------------------------------------------------------------------|
| $\gamma$     | Constant ( $\sim 0.7$ )                                                   |
| $\epsilon_b$ | Porosity of the adsorbent bed                                             |
| $\mu$        | Viscosity of the carrier gas, g/cm min                                    |
| $\rho$       | Density of the carrier gas, $\text{g/cm}^3$                               |
| $\rho_b$     | Bulk density of the adsorbent bed ( $M/V_{\text{bed}}$ ), $\text{g/cm}^3$ |
| $\rho_p$     | Density of an adsorbent particle, $\text{g/cm}^3$                         |

## Acknowledgement

An abridged version of this work was presented at the CHEMCON 2007 conference held at Heritage Institute of Technology, Kolkata, India during 27–30 December, 2007.

## References

1. Walker, G.M. and Weatherley, L.R., "Adsorption of Acid Dyes on to Granular Activated Carbon in Fixed Beds", *Water Res.*, **31**, pp. 2093–2101 (1997).
2. Han, R., Zou, W., Li, H., Li, Y. and Shi, J., "Copper(II) and Lead(II) Removal from Aqueous Solution in Fixed-bed Columns by Manganese Oxide Coated Zeolite", *J. Hazard. Mater.*, **B137**, pp. 934–942 (2006).
3. Mak, V.W.Y., Chatzis, I., Hudgins, R.R. and Fayed, M.E., "Removal of VOCs by Sorption in Fixed Beds of Popcorn", *Can. J. Chem. Eng.*, **82**, pp. 194–197 (2004).
4. Goud, V.V., Mohanty, K., Rao, M.S. and Jayakumar, N.S., "Phenol Removal from Aqueous Solutions by Tamarind Nutshell Activated Carbon: Batch and Column Studies", *Chem. Eng. Technol.*, **28**, pp. 814–821 (2005).
5. Mohanty, K., Naidu, J.T., Meikap, B.C. and Biswas, M.N. "Removal of Crystal Violet from Wastewater by Activated Carbons Prepared from Rice Husk", *Ind. Eng. Chem. Res.*, **45**, pp. 5165–5171 (2006).
6. Ho, Y.S. and McKay, G., "The Kinetics of Sorption of Divalent Metal Ions onto Sphagnum Moss Peat", *Wat. Res.*, **34**, pp. 735–742 (2000).
7. Allen, S.J., Gan, Q., Matthews, R. and Johnson, P.A., "Kinetic Modeling of the Adsorption of Basic Dyes by Kudzu", *J. Coll. Interf. Sci.*, **286**, pp. 101–109 (2005).
8. Manju, G.N., Raji, C. and Anirudhan, T.S., "Evaluation of Coconut Husk Carbon for the Removal of Arsenic from Water", *Wat. Res.*, **32**, pp. 3062–3070 (1998).
9. Hamdaoui, O., "Dynamic Sorption of Methylene Blue by Cedar Sawdust and Crushed Brick in Fixed Bed Columns", *J. Hazard. Mater.* **B138**, pp. 293–303 (2006).
10. Gupta, V.K., Mohan, D. and Singh, K.P. "Removal of 2-Aminophenol Using Novel Adsorbents", *Ind. Eng. Chem. Res.*, **45**, pp. 1113–1122, (2006).

11. Gupta, V.K., Srivastava, S.K., Mohan, D. and Sharma, S., "Design Parameters for Fixed Bed Reactors of Activated Carbon Developed from Fertilizer Waste for the Removal of Some Heavy Metal Ions", *Waste Manag.*, **17**, pp. 517–522 (1997).
12. Srivastava, S.K., Tyagi, R. and Pant, N., "Adsorption of Heavy Metal Ions on Carbonaceous Material Developed from the Waste Slurry Generated in Local Fertilizer Plants", *Wat. Res.*, **23**, pp. 1161–1165 (1989).
13. Ajmal, M., Rao, R.A.K. and Khan, M.A., "Adsorption of Copper from Aqueous Solution on *Brassica cumpestris* (Mustard Oil Cake)", *J. Hazard. Mater.*, **B122**, pp. 177–183 (2005).
14. Gupta, V.K. and Sharma, S., "Removal of Cadmium and Zinc from Aqueous Solutions Using Red Mud", *Environ. Sci. Technol.*, **36**, pp. 3612–3617 (2002).
15. Gupta, V.K., Gupta, M. and Sharma, S., "Process Development for the Removal of Lead and Chromium from Aqueous Solutions Using Red Mud – an Aluminum Industry Waste", *Wat. Res.*, **35**, pp. 1125–1134 (2001).
16. Ahmaruzzaman, M. and Sharma, D.K., "Adsorption of Phenols from Wastewater", *J. Coll. Interf. Sci.*, **287**, pp. 14–24 (2005).
17. Purkait, M.K., DasGupta, S. and De, S., "Adsorption of Eosin Dye on Activated Carbon and its Surfactant Based Desorption", *J. Env. Manag.*, **76**, pp. 135–142 (2005).
18. Amin, M.N., Kaneco, S., Kitagawa, T., Begum, A., Katsumata, H., Suzuki, T. and Ohta, K., "Removal of Arsenic in Aqueous Solutions by Adsorption into Waste Rice Husk", *Ind. Eng. Chem. Res.*, **45**, pp. 8105–8110 (2006).
19. Yeddou, N. and Bensmaili, A., "Kinetic Models for the Sorption of Dye from Aqueous Solution by Clay–Wood Sawdust Mixture", *Desalination*, **185**, pp. 499–508 (2005).
20. Maji, S.K., Pal, A., Pal, T. and Adak, A., "Modeling and Fixed Bed Column Adsorption of As(III) on Laterite Soil", *Sep. Purif. Technol.*, **56**, pp. 284–290 (2007).
21. Gupta, V.K., "Equilibrium Uptake, Sorption Dynamics, Process Development, and Column Operations for the Removal of Copper and Nickel from Aqueous Solution and Wastewater Using Activated Slag, a Low-Cost Adsorbent", *Ind. Eng. Chem. Res.*, **37**, pp. 192–202 (1998).
22. Mohanty, K., Jha, M., Meikap, B.C. and Biswas, M.N., "Removal of Chromium(VI) from Dilute Aqueous Solutions by Activated Carbon Developed from Terminalia Arjuna Nuts Activated with Zinc Chloride", *Chem. Eng. Sci.*, **60**, pp. 3049–3059 (2005).
23. Sharma, A. and Bhattacharyya, K.G., "Azadirachta Indica (Neem) Leaf Powder as a Biosorbent for the Removal of Cd(II) from Aqueous Medium", *J. Hazard. Mater.*, **B125**, pp. 102–112 (2005).
24. Lagergren, S., "Zur Theorie Der Sogenannten Adsorption Gelöster Stoffe", *Kungliga Svenska Vetenskapsakademiens Handlingar*, **24**, pp. 1–39 (1898).
25. Ho, Y.S. and McKay, G., "Pseudo-second Order Model for Sorption Processes", *Process Biochem.*, **34**, pp. 451–465 (1999).
26. Azizian, S., "Kinetic Models of Sorption: A Theoretical Analysis", *J. Coll. Interf. Sci.*, **276**, pp. 47–52 (2004).
27. Tien, C., *Adsorption Calculations and Modeling*, Butterworth-Heinemann, Boston, MA (1994).
28. Tien, C., "Remarks on Adsorption Manuscripts Received and Declined: An Editorial", *Sep. Purif. Technol.*, **54**, pp. 277–278 (2007).
29. Tiwari, D., Kim, H.-U. and Lee, S.-M., "Removal Behavior of Sericite for Cu(II) and Pb(II) from Aqueous Solutions: Batch and Column Studies", *Sep. Purif. Technol.*, **57**, pp. 11–16 (2007).
30. Kumar, G.P., Kumar, P.A., Chakraborty, S. and Ray, M., "Uptake and Desorption of Copper Ion Using Functionalized Polymer Coated Silica Gel in Aqueous Environment", *Sep. Purif. Technol.*, **57**, pp. 47–56 (2007).
31. Gezici, O., Kara, H., Ayar, A. and Topkafa, M., "Sorption Behavior of Cu(II) ions on Insolubilized Humic Acid under Acidic Conditions: An Application of Scratchard Plot Analysis in Evaluating the pH Dependence of Specific and Nonspecific Bindings", *Sep. Purif. Technol.*, **55**, pp. 132–139 (2007).

32. Weng, C.-H., Tsai, C.-Z., Chu, S.-H. and Sharma, Y.C., "Adsorption Characteristics of Copper(II) onto Spent Activated Clay", *Sep. Purif. Technol.*, **54**, pp. 187–197 (2007).
33. van der Sman, R.G.M., "Simple Model for Estimating Heat and Mass Transfer in Regular-shaped Foods", *J. Food Engineering*, **60**, pp. 383–390 (2003).
34. Mittal, A., Kataria, T., Das, G.K. and Chatterjee, S.G., "Evaporative Cooling of Water in a Small Vessel Under Varying Ambient Humidity", *Int. J. Green Energy*, **3**, pp. 347–368 (2006).
35. Chatterjee, S.G., "A Simple Description of Transient Moisture Diffusion in Paper", *Indian Chemical Engineer*, **50**, pp. 34–46 (2008).
36. Richardson, J.F., Harker, J.H. and Backhurst, J.R., *Coulson and Richardson's Chemical Engineering*, Vol. 2, 5<sup>th</sup> edn, Butterworth-Heinemann, London (2002).

## Appendix

In this appendix, a theoretical justification is provided for Equations (6) and (26), assuming physical adsorption. Consider a batch adsorption system in which the dosage of the adsorbent, which consists of spherical particles of radius  $a_p$ , is  $C_{ads}$ . Let  $c$  and  $q$  be the adsorbate concentrations in the fluid and solid phases, respectively, at any time  $t$  after the start of the adsorption process, and let  $c_0$  and  $q_0$  be the initial values of  $c$  and  $q$  at  $t=0$ , respectively. An overall mass balance yields

$$c = c_0 - C_{ads}(q - q_0) \quad (A1)$$

In the adsorption literature, there is a well-known approximation known as the Glueckauf or linear-driving-force model that is used for describing the uptake of an adsorbate by an adsorbent pellet [27]. The uptake rate has the form

$$\frac{dq}{dt} = k_p[q_s(t) - q] \quad (A2)$$

where  $q_s$  is the solid-phase concentration of the adsorbate at the pellet's surface and  $k_p$  is an intraparticle mass-transfer coefficient, which for a sphere of radius  $a_p$ , is given by [27]

$$k_p = \frac{15D_e}{a_p^2} \quad (A3)$$

Here,  $D_e$  is the effective diffusion coefficient of the adsorbate in the pellet. The linear-driving-force approximation has also been used for describing evaporative cooling [33, 34] and moisture diffusion in Chatterjee [35]. A mass balance for adsorbate around a single adsorbent pellet yields

$$\frac{dq}{dt} = \frac{3k_f}{\rho_p a_p} [c - c_s(t)] \quad (A4)$$

where  $k_f$  is the fluid-phase mass-transfer coefficient,  $\rho_p$  is the pellet density and  $c_s$  is the fluid-phase concentration of the adsorbate at the surface of the pellet. Assuming equilibrium at the pellet's surface and a linear isotherm, one has

$$q_s = K_{eq}c_s \quad (A5)$$

where  $K_{\text{eq}}$  is the slope of the isotherm. From Equations (A1), (A2), (A4) and (A5), it may be shown that

$$\frac{dq}{dt} = k_1[q_e - q] \quad (\text{A6})$$

which is the first-order kinetic or Lagergren model [i.e. Equation (6)]. Here,  $q_e$  is the maximum or ultimate concentration of the adsorbate in the solid, which is given by

$$q_e = \frac{c_0 + C_{\text{ads}}q_0}{C_{\text{ads}} + 1/K_{\text{eq}}} \quad (\text{A7})$$

and

$$k_1 = k_{\text{eff}}(K_{\text{eq}}C_{\text{ads}} + 1) \quad (\text{A8})$$

Also,  $k_{\text{eff}}$  is an effective or overall mass-transfer coefficient defined by

$$k_{\text{eff}} = \frac{k_p k'_f}{k_p K_{\text{eq}} + k'_f} \quad (\text{A9})$$

where

$$k'_f = \frac{3k_f}{\rho_p a_p} \quad (\text{A10})$$

From Equations (A9) and (A10) it follows that, as  $k_f \rightarrow \infty$ , one has

$$k_{\text{eff}}(k_f \rightarrow \infty) \rightarrow k_p \quad (\text{A11})$$

To generalize from a batch adsorption system to a fixed bed of adsorbent particles, it is proposed that  $D_e$  be replaced by  $D_L$  (axial dispersion coefficient of the adsorbate in the bed),  $a_p$  by  $L$ ,  $C_{\text{ads}}$  by  $\rho_b/\varepsilon_b$ , and  $K_{\text{eq}}$  by  $q_e/c_{\text{in}}$ . A dimensionless group that is expected to play a role in fixed-bed adsorption is  $L/d_p$  where  $d_p$  is the particle diameter. Equations (A3), (A8) and (A11) then suggest the following correlation for fixed-bed adsorption:

$$\frac{k_1 L^2}{D_L} = A \left( \frac{n_{\text{stoic}}}{\varepsilon_b} + 1 \right)^B \left( \frac{L}{d_p} \right)^C \quad (\text{A12})$$

where  $A$ ,  $B$  and  $C$  are constants that have to be determined experimentally. The factor 1 on the right-hand side of Equation (A12) represents the case of pure mass transfer (i.e. no adsorption). For beds of particles, the axial dispersion coefficient  $D_L$  can be obtained from the expressions given below [36].

For gases:

$$\begin{aligned}
 D_L &= \gamma D \approx 0.7D \text{ for } \text{Re}_p < 1 \\
 &= \frac{1}{2} \frac{u_s d_p}{\varepsilon_b} \text{ for } \text{Re}_p > 10 \\
 &= \gamma D + \frac{1}{2} \frac{u_s d_p}{\varepsilon_b} \text{ for } 1 < \text{Re}_p < 10
 \end{aligned}
 \tag{A13}$$

For liquids:

$$D_L \approx \frac{u_s d_p}{\varepsilon_b} \tag{A14}$$

In Equation (A13),  $D$  is the molecular diffusion coefficient of the adsorbate species in the carrier gas and  $\text{Re}_p$  is the particle Reynolds number defined by

$$\text{Re}_p = \frac{u_s d_p \rho}{\mu} \tag{A15}$$

where  $\rho$  and  $\mu$  are the density and viscosity of the gas, respectively.